EPSR-test-ex80 dockato: water onto water, answers from standard input

Program(s): dockato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

`dockato` [1] builds larger molecules from `.ato` building blocks: copies of the first molecule of a docking file are bonded onto free sites of a docked file, removing one atom of each to form the new bond. This case chains water molecules and reads the docked-site parameters from standard input.

Code exercised

Fortran (original, fortran_src/):

- `dockato.f`: `dockato` (driver, parameters from command line or stdin), `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`
- `dockato.f`: `test_overlap`, `bond_molecule` (ROT/DIH groups), `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`
- `ato_manipulation_rev4.f`: `get_direct_bonds`, `get_rotgrp`, `centreofmass`, `ftype`; `ato_read_and_save.f`: `moloutc`

Rust (port, rust/epsr-core/src/):

- `tools/dockato.rs`: `run`, `rotatem`, `random_rotatem`, `search_unsaturated`, `find_valence`, `test_overlap`
- `tools/dockato.rs`: `bond_molecule`, `replace_references`, `add_neighbours`, `remove_atoms`, `clean_atom_types`, `get_direct_bonds`, `get_rotgrp`
- `ato.rs`: `State::moloutc` (ROT/DIH groups), `State::ftype`; `rng.rs`: `Ran1::ran1`

Physical system

The building block `WaterMolecule.ato` of EPSR-test: one water molecule (O1, H1, H1; O-H 0.976 Å, H-H 1.54 Å, SPC/E Lennard-Jones and charges [2]) in a 14.42 Å box at 300 K. Atom `atomd` of the new molecule is placed on `atomv`, the molecule is rotated so that its `atomdr` neighbour lies on the `atomvr` neighbour of `atomv` (equal bond lengths required), an optional second (dihedral) rotation about the new bond uses `atomdrr`/`atomvrr` (random angle in [dminrr, dmaxrr] from `ran1`, the Park-Miller generator with Bays-Durham shuffle [3,4], or a fixed dihedral that creates a DIH group when dminrr = dmaxrr), overlaps are tested with a distance factor, `atomd` and `atomvr` are removed and the new rotational group, bonds and atom types are updated. Here the docking H1 goes onto the docked O1 and the docking O1 onto the docked H1, so both H atoms are removed and an O-O bond is formed: a chain of water units joined by O-O bonds (a test structure, not a real molecule), with new ROT groups and the box rescaled.

Inputs

Docking parameters on the command line (`H1 O1 H1`, radmax 10); run 1: `O1 H1 H1`, dihedral 30-90°, overlap factor 1.0 and output name from `dock_random.stdin`, 3 molecules; run 2: `dock_yes.stdin` with -60-60°, `yes` (overlaps allowed), a blank line before the output name (re-prompt) and the name given with `.ato`, 6 molecules.

How the accuracy was verified

`commands.txt` (2 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Each run writes one new `.ato` file (coordinates, bonds, rotational groups, atom types and the random-number state), compared byte for byte with the stored Fortran output. The overlap test is active: `waterchain.ato` holds 5 atoms (3 + 1 per docking), i.e. 2 of 3 placements accepted, with 2 ROT groups and box side 21.90 Å; `waterchain_yes.ato` has 9 atoms, 6 ROT groups (all 6 dockings) and box side 36.72 Å.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).

[3] S. K. Park and K. W. Miller, "Random number generators: good ones are hard to find", Commun. ACM 31, 1192-1201 (1988).

[4] W. H. Press, S. A. Teukolsky, W. T. Vetterling and B. P. Flannery, Numerical Recipes in Fortran 77, 2nd ed., Cambridge University Press (1992).

[5] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.